Member 1 – Dataset Collection + Inventory
- Download and inspect the Kaggle PDF dataset.
- Collect a representative set of college material from different courses.
- Include different types wherever available: slides, notes, books, scanned notes, handwritten material, etc.
- Prepare an inventory containing:
  file/source, subject, file type, number of pages, file size, selectable/scanned text, etc.
- Compare the diversity of Kaggle documents vs college-specific documents.

Member 2 – PDF/Text Extraction Analysis
- Extract text from a representative sample from BOTH sources.
- Check:
  missing text
  broken text
  reading order
  headers/footers
  special symbols
  equations
  tables
  repeated content
- Calculate basic document statistics such as:
  pages/document
  words/document
  characters/document
  amount of extractable text
- Compare extraction quality between Kaggle and college documents.

Member 3 – Scanned/OCR + Data Quality Analysis
- First understand what is OCR and why is it used.
- Identify scanned/image-based documents.
- Apply OCR on representative samples.
- Check OCR errors caused by:
  handwriting
  low resolution
  skew
  equations
  tables
  diagrams
- Try basic preprocessing where required.
- Compare OCR quality before and after preprocessing.
- Categorize documents as easy/moderate/difficult to process.

Member 4 – Document Structure + Chunking Analysis
- First understand what is chunking and why it is used.
- Analyze headings, subheadings, paragraphs, bullet points, tables and sections.
- Try:
  fixed-size chunking
  paragraph-based chunking
  heading/section-based chunking
- Compare:
  chunk sizes
  information loss

  overlap/repetition
  preservation of context
- Give examples of good and bad chunks from actual documents.

Member 5 – Combined Dataset Analysis + Quiz Readiness
- Compare Kaggle vs college material in terms of:
  document type
  length
  structure
  extraction quality
  subject specificity
  noise
- Identify which material is actually useful for quiz generation.
- Create 15–20 questions from different document types.
- Record for each:
  source passage
  question type
  answer
  difficulty
- Define what a good generated question should satisfy:
  correctness
  relevance
  answerability
  non-duplication
  appropriate difficulty

By next lab, you should have:
- Final dataset/source inventory
- Dataset statistics
- Text-extraction analysis
- OCR analysis
- Document-quality analysis
- Chunking comparison
- Kaggle vs college-material comparison
- Important plots/tables
- Clear observations and conclusions